

2026.9.24

Last time

THM Every finitely generated abelian group is of the form 意义: G 可拆成若干无限循环群和若干有限循环群的直积.
 $G \cong \mathbb{Z}^r \times \mathbb{Z}/n_1\mathbb{Z} \times \dots \times \mathbb{Z}/n_k\mathbb{Z}$, $r \geq 0, 2 \leq n_1 | n_2 | \dots | n_k$ uniquely determined. 而循环群更易研究.

THM (Chinese remainder theorem. Special case)
if $m, n \in \mathbb{Z}_{\geq 2}$, $\gcd(m, n) = 1$, then $\mathbb{Z}/mn\mathbb{Z} \cong \mathbb{Z}/m\mathbb{Z} \times \mathbb{Z}/n\mathbb{Z}$.

Pf. Consider $\phi: \mathbb{Z}/mn\mathbb{Z} \rightarrow \mathbb{Z}/m\mathbb{Z} \times \mathbb{Z}/n\mathbb{Z}$
 $a \bmod mn \mapsto (a \bmod m, a \bmod n)$
group homomorphism $\ker \phi = \{a \bmod mn \mid a \equiv 0 \pmod m, a \equiv 0 \pmod n\} = \{0\}$ $\xrightarrow{\gcd(m, n)=1}$
 $\Rightarrow \phi$ injective but $\#\mathbb{Z}/mn\mathbb{Z} = \#(\mathbb{Z}/m\mathbb{Z} \times \mathbb{Z}/n\mathbb{Z}) \Rightarrow \phi$ isomorphism. $\square$

COR Every f.g. abelian group is of the form $G \cong \mathbb{Z}^r \times (\mathbb{Z}/p_1^{f_{11}}\mathbb{Z} \times \dots \times \mathbb{Z}/p_1^{f_{1s_1}}\mathbb{Z}) \times (\mathbb{Z}/p_2^{f_{21}}\mathbb{Z} \times \dots \times \mathbb{Z}/p_2^{f_{2s_2}}\mathbb{Z}) \times \dots$
 r, p_i, f_{ij} are uniquely determined. 例子: 12阶交换群. $12 = 2^2 \times 3$.
 $2 = 2 = 1+1 \rightarrow \mathbb{Z}/4\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z} \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z} \cong \mathbb{Z}/12\mathbb{Z} \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/6\mathbb{Z}$.
 $\uparrow$ 则12阶交换群只有以上2种.

Conclusion can classify all finite abelian groups of any order. 对 $|G| = \prod p_i^{a_i}$, 将 a_i 写成正整数之和 (无序)

Pf comes later. abelian groups = $\mathbb{Z}$ -modules 主理想整环 拆成素数幂, 再按素数归类.

Will give a more general classification of finitely generated modules over PID (principal ideal domain).

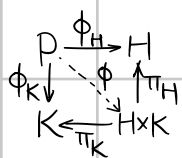
§5. Direct product, group actions, semi-direct products

Recall $G = H \times K \ni (h, k)$
 $(h, k) \cdot (h', k') = (hh', kk')$
 $e_G = (e_H, e_K)$
 $(h, k)^{-1} = (h^{-1}, k^{-1})$

there are natural embeddings $H \hookrightarrow G, K \hookrightarrow G$
 $h \mapsto (h, e_K), k \mapsto (e_H, k)$
injective homomorphism realizing H, K as normal subgroups of G . H, K 均被实现为正规子群, 两“分量”互不影响. 但一般的群 G 不一定具有这么好的解耦结构. 这是引入半直积的动机之一.

$G/H \cong K, G/K \cong H$ there are natural projections $\pi_H: G \rightarrow H, \pi_K: G \rightarrow K$. 表达更复杂的结构.

REM 1). (universal property). $\forall$ group P , homomorphism $\phi_H: P \rightarrow H, \phi_K: P \rightarrow K$,
 $\exists$ unique homomorphism $\phi: P \rightarrow G = H \times K$ 即给定一个映入 $H \times K$ 的 hom, 等价于分别给定一个映入 H 和 K 的 hom.



making the diagram commute.

(exercise: universal property for the quotient group G/N)

2) Can also define infinite direct products $\prod_{i \in I} G_i$ and direct sums.

PROP (Recognizing direct products). Let G be a group with two normal subgroups $H, K \triangleleft G$, st. $HNK = \{e\}$

Then $HK \cong H \times K$.

Pf. $H, K \triangleleft G \Rightarrow HK \triangleleft G$. Consider $\phi: H \times K \rightarrow HK$
 $(h, k) \mapsto hk$

ϕ is a homomorphism:

$$\begin{aligned}
 \phi((h_1, k_1) \cdot (h_2, k_2)) &= \phi(h_1, k_1) \cdot \phi(h_2, k_2) \Leftrightarrow h_1 h_2 k_1 k_2 = h_1 k_1 h_2 k_2 \Leftrightarrow h_2 k_1 = k_1 h_2 \Leftrightarrow \underbrace{h_2 k_1 h_2^{-1}}_{\substack{\downarrow \\ K}} \underbrace{k_1^{-1}}_{\substack{\downarrow \\ H}} = e \\
 &\quad \text{But } h_2 k_1 h_2^{-1} k_1^{-1} \in HNK = \{e\}
 \end{aligned}$$

ϕ is surjective: obvious

$$\ker \phi = \{(h, k) \in H \times K \mid hk = e\} \Rightarrow \ker \phi = \{(e, e)\} \Rightarrow \phi \text{ is an isomorphism. } \square$$

★ 群作用 — 群概念的来源.

Group actions (Motivating examples $D_n, S_n, GL_n, \dots$) 为了把一个群作用在一个集合上, 可用这作用去研究群的结构.

DEF Let G be a group and X a set.

A left G -action on X $G \curvearrowright X$ is a map $G \times X \rightarrow X$, st.
 $(g, x) \mapsto g \cdot x$

1) $\forall g, h \in G, x \in X$, we have $g \cdot (h \cdot x) = (gh) \cdot x$ } 与群结构相容

2) $\forall x \in X, e \cdot x = x$.

LEM This implies $\forall g \in G, \phi_g: X \rightarrow X$ is a bijection with inverse $\phi_g^{-1} = \phi_{g^{-1}}$.

eg 1) $S_n \curvearrowright X = \{1, 2, \dots, n\}$ $\sigma(\tau(i)) = (\sigma \cdot \tau)(i)$.

2) $D_n \curvearrowright X = \text{regular } n\text{-gon}$.

3) If $G \curvearrowright X$ $H \leq G$, then get $H \curvearrowright X$ by restriction.

4) G -actions on G itself

• 左平移 $\forall g \in G, \ell_g: G \rightarrow G$ $\ell_g(x) := gx$.

• 右平移 $\forall g \in G, \gamma_g: G \rightarrow G$ $\gamma_g(x) := xg^{-1}$.

why g^{-1} : because we want $\Gamma g \circ \Gamma h(x) = \Gamma gh(x)$.

• conjugation action. $\forall g \in G$, $\text{Ad}_g: G \rightarrow G$ $\text{Ad}_g(x) = gxg^{-1}$ $G \xrightarrow{\text{Ad}} G$

This behaves better.

$$\text{Ad}_g(xy) = gxyg^{-1} = gxg^{-1}gyg^{-1} = \text{Ad}_g(x)\text{Ad}_g(y).$$

$\Rightarrow \forall g \in G$, $\text{Ad}_g: G \xrightarrow{\sim} G$ is a self-isomorphism.
 // automorphism

DEF A right G -action on X $X \curvearrowright G$ is a map $X \times G \rightarrow X$,
 $(x, g) \mapsto x \cdot g$

st. $(x \cdot g) \cdot h = x \cdot (gh)$ and $x \cdot e = x$.

PROP Let G be a group action as a set X

Then there is a natural homomorphism $\Phi: G \rightarrow S_X = \text{symmetric/permutation group of } X$.

In fact, giving a $G \curvearrowright X$
 $g \mapsto \Phi_g: X \rightarrow X$
 $x \mapsto gx$
 记录群作用信息的方法

is equivalent to giving such a homomorphism.

PF Check $\Phi_g \circ \Phi_h = \Phi_{gh}$. $\forall x \in X$, $\Phi_g \circ \Phi_h(x) = \Phi_g(hx) = g(hx) = (gh)x = \Phi_{gh}(x)$. $\square$

DEF If Φ is injective, we say that the G -action is faithful.
 忠实的 $\rightarrow$ 即从操作反推出群元素.

(ie. $\ker \Phi = \{e\}$, meaning no nontrivial element of G fixes all elements of X).

If Φ is trivial, ie. Φ_g is id, $\forall g \in G$, we say that the G -action is trivial.

THM (Cayley's theorem) ¹⁸⁵⁴ Every group G is isomorphic to a subgroup of some symmetric group.

if $\#G = n$, then G is isomorphic to a subgroup of S_n .

PF Left translation action gives $\Phi: G \hookrightarrow S_G$.
 $g \mapsto \ell_g$ $\square$

DEF $\text{Aut}(G) := \{\text{automorphism } G \cong G\}$ form a group.

• group operation = composition

• identity is id: $G \cong G$.

we have $\text{Aut}(G) < S_G$.

eg. The conjugation action $G \xrightarrow{\text{Ad}} G$ induces a natural homomorphism.

$$\text{Ad}: G \rightarrow \text{Aut}(G) < S_G$$

与被作用的集合的结构相容
此处是群

$$g \mapsto \text{Ad}_g: G \cong G$$

$$x \mapsto g \times g^{-1}$$

REM. A slight generalization.

X itself is a group. $G \curvearrowright X$ preserving group structures,

$$\text{i.e. } \forall g \in G, \phi_g: X \xrightarrow{\sim} X \text{ is an automorphism.}$$

$$x \mapsto gx$$

$$\text{equivalently, } \Phi: G \rightarrow \text{Aut}(X) < S_X$$

半直积

$$g \mapsto \Phi_g$$

Semi-direct products. more ways to build larger groups

$$H, N < G. \quad N \text{ normal but } H \text{ not.} \quad H \cap N = \{e\}.$$

$$\text{Consider } NH < G. \quad n_1 h_1 \cdot n_2 h_2 = n_1 h_1 n_2 h_1^{-1} h_2$$

需记录 h 对 N 中所有元素的作用的信息.

DEF. Let N and H be groups and $\Phi: H \rightarrow \text{Aut}(N)$ a homomorphism. $\forall h \in H$, with $\Phi_h := \Phi(h): N \xrightarrow{\sim} N$

$$h \mapsto \Phi_h$$

Defines the semi-direct product $N \rtimes H$ or $N \rtimes_{\Phi} H$ to be

$$N \rtimes H = \{(n, h) \mid n \in N, h \in H\} \quad (n_1, h_1) \cdot (n_2, h_2) := (n_1 \Phi_{h_1}(n_2), h_1 h_2)$$

易证 $N \rtimes H$ 是群.

$$N \hookrightarrow N \rtimes H \quad H \hookrightarrow N \rtimes H$$

we have natural embeddings

$$n \mapsto (n, e_H)$$

$$h \mapsto (e_N, h)$$

$$e_{N \rtimes H} = (e_N, e_H)$$

$$(n, h)^{-1} = (\Phi_h^{-1}(n^{-1}), h^{-1})$$

and a natural projection $\pi: N \rtimes H \rightarrow H$
 $(n, h) \mapsto h$ with $N \cong \ker \pi$ normal subgroup.

PROP. (Recognizing semi-direct products) Let G be a group $H < G, N < G$, st. $H \cap N = \{e\}$.

$$\text{Then } NH \cong N \rtimes_{\text{Ad}} H \text{ where } \begin{aligned} \text{Ad}: H &\rightarrow \text{Aut}(N) \\ h &\mapsto \text{Ad}_h: N \xrightarrow{\sim} N \\ n &\mapsto h n h^{-1} \end{aligned}$$

$$\text{pf. Consider } \Psi: N \rtimes_{\text{Ad}} H \rightarrow NH$$

$$(n, h) \mapsto nh$$

Ψ is a homomorphism. Ψ surjective and $\ker \Psi = H \cap N = \{e\} \Rightarrow \Psi$ isomorphism. $\square$

$$\text{eg. } 1) \mathbb{Z}/2\mathbb{Z} < \mathbb{Z}/4\mathbb{Z} \quad (\mathbb{Z}/4\mathbb{Z})/(\mathbb{Z}/2\mathbb{Z}) = \mathbb{Z}/2\mathbb{Z}$$

but $\mathbb{Z}/4\mathbb{Z}$ not a (semi-)direct product of $\mathbb{Z}/2\mathbb{Z}$ and $\mathbb{Z}/2\mathbb{Z}$.

$$2) \{\pm 1\} \curvearrowright \mathbb{Z}/n\mathbb{Z} \text{ by multiplication} \quad \mathbb{Z}/n\mathbb{Z} \rtimes \{\pm 1\} \cong D_{2n}$$

